

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

Advanced Math

06

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**A**

A. $4x^2 + 21x + 33$

Choice A is correct. The given expression can be rewritten as $(2x + 5)^2 + (-1)(x - 2) + 2(x + 3)$. Applying the distributive property, the expression $(-1)(x - 2) + 2(x + 3)$ can be rewritten as $-1(x) + (-1)(-2) + 2(x) + 2(3)$, or $-x + 2 + 2x + 6$. Adding like terms yields $x + 8$. Substituting $x + 8$ for $(-1)(x - 2) + 2(x + 3)$ in the given expression yields $(2x + 5)^2 + x + 8$. By the rules of exponents, the expression $(2x + 5)^2$ is equivalent to $(2x + 5)(2x + 5)$. Applying the distributive property, this expression can be rewritten as $2x(2x) + 2x(5) + 5(2x) + 5(5)$, or $4x^2 + 10x + 10x + 25$. Adding like terms gives $4x^2 + 20x + 25$. Substituting $4x^2 + 20x + 25$ for $(2x + 5)^2$ in the rewritten expression yields $4x^2 + 20x + 25 + x + 8$, and adding like terms yields $4x^2 + 21x + 33$.

Choices B, C, and D are incorrect. Choices C and D may result from rewriting the expression $(2x + 5)^2$ as $4x^2 + 25$, instead of as $4x^2 + 20x + 25$. Choices B and C may result from rewriting the expression $-(x - 2)$ as $-x - 2$, instead of $-x + 2$.

Q2**B****B.**

x	y
-6	-3
-5	-3
4	-3

Choice B is correct. It's given that $fx = (x + 6)(x + 5)(x - 4)$ and $y = f(x) - 3$. Substituting $(x + 6)(x + 5)(x - 4)$ for fx in the equation $y = f(x) - 3$ yields $y = (x + 6)(x + 5)(x - 4) - 3$. Substituting -6 for x in this equation yields $y = (-6 + 6)(-6 + 5)(-6 - 4) - 3$, or $y = -3$. Substituting -5 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (-5 + 6)(-5 + 5)(-5 - 4) - 3$, or $y = -3$. Substituting 4 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (4 + 6)(4 + 5)(4 - 4) - 3$, or $y = -3$. Therefore, when $x = -6$ then $y = -3$, when $x = -5$ then $y = -3$, and when $x = 4$ then $y = -3$. Thus, the table of values in choice B represents $y = f(x) - 3$.

Choice A is incorrect. This table represents $y = x - 3$ rather than $y = f(x) - 3$.

Choice C is incorrect. This table represents $y = x + 3$ rather than $y = f(x) - 3$.

Choice D is incorrect. This table represents $y = fx + 3$ rather than $y = f(x) - 3$.

Q3**4**

The correct answer is 4. The left-hand side of the given equation can be factored as $7x + 8x - 4$. Therefore, the given equation, $7x^2 - 20x - 32 = 0$, can be written as $7x + 8x - 4 = 0$. Applying the zero product property to this equation yields $7x + 8 = 0$ and $x - 4 = 0$. Subtracting 8 from both sides of the equation $7x + 8 = 0$ yields $7x = -8$. Dividing both sides of this equation by 7 yields $x = -\frac{8}{7}$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the two solutions to the given equation, $7x^2 - 20x - 32 = 0$, are $-\frac{8}{7}$ and 4. It follows that 4 is the positive solution to the given equation.

Q4**9**

The correct answer is 9. The given expression can be rewritten as $5x^3 - 3 + -1-4x^3 + 8$. By applying the distributive property, this expression can be rewritten as $5x^3 - 3 + 4x^3 + -8$, which is equivalent to $5x^3 + 4x^3 + -3 + -8$. Adding like terms in this expression yields $9x^3 - 11$. Since it's given that $5x^3 - 3 - -4x^3 + 8$ is equivalent to $bx^3 - 11$, it follows that $9x^3 - 11$ is equivalent to $bx^3 - 11$. Therefore, the coefficients of x^3 in these two expressions must be equivalent, and the value of b must be 9.

Q5**D****D. 9**

Choice D is correct. It's given that the equation $h = -4.9t^2 + 7t + 9$ represents this situation, where h is the height, in meters, of the object t seconds after it is kicked. It follows that the height, in meters, from which the object was kicked is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 7t + 9$ yields $h = -4.9(0)^2 + 7(0) + 9$, or $h = 9$. Therefore, the object was kicked from a height of 9 meters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6**D****D. Zero**

Choice D is correct. Since the square of a real number is never negative, the given equation isn't true for any real value of x . Therefore, the given equation has zero distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**D**

D. $(x+4)^{\frac{1}{2}}$

Choice D is correct. The given expression can also be written as $(x^2+8x+16)^{\frac{1}{4}}$. The trinomial $x^2+8x+16$ can be rewritten in factored form as $(x+4)^2$. Thus, the entire expression can be rewritten as $((x+4)^2)^{\frac{1}{4}}$. Simplifying the exponents yields $(x+4)^{\frac{1}{2}}$.

Choices A, B, and C are incorrect and may result from errors made when simplifying the exponents in the expression $((x+4)^2)^{\frac{1}{4}}$.

Q8**A**

A. $f(0)$

Choice A is correct. A y-intercept is the point in the xy-plane where the graph of the function crosses the y-axis, which is where $x=0$. It's given that the y-coordinate of the y-intercept of the graph of function f is c . It follows that the coordinate pair representing the y-intercept must be $(0,c)$. Therefore, c must equal $f(0)$.

Choices B, C, and D are incorrect because $f(1)$, $f(2)$, and $f(3)$ would represent the y-value of the coordinate where $x=1$, $x=2$, and $x=3$, respectively.

Q9**C****C.** 8

Choice C is correct. The left-hand side of the given equation can be factored as $5x + 3x - 8$. Therefore, the given equation, $5x^2 - 37x - 24 = 0$, can be written as $5x + 3x - 8 = 0$. Applying the zero product property to this equation yields $5x + 3 = 0$ and $x - 8 = 0$. Subtracting 3 from both sides of the equation $5x + 3 = 0$ yields $5x = -3$. Dividing both sides of this equation by 5 yields $x = -\frac{3}{5}$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Therefore, the two solutions to the given equation, $5x^2 - 37x - 24 = 0$, are $-\frac{3}{5}$ and 8. It follows that 8 is the positive solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**B****B.** $\frac{8x - 3}{2}$

Choice B is correct. The given expression has a common factor of 2 in the denominator, so the expression can be rewritten as $\frac{8xx - 7 - 3x - 7}{2x - 7}$. The three terms in this expression have a common factor of $x - 7$. Since it's given that $x > 7$, x can't be equal to 7, which means $x - 7$ can't be equal to 0. Therefore, each term in the expression, $\frac{8xx - 7 - 3x - 7}{2x - 7}$, can be divided by $x - 7$, which gives $\frac{8x - 3}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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